LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	21
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Turn forces in a turn, rate of turn steady level coordinated turn and control in a turn.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Turn forces in a turn, rate of turn steady level coordinated turn and control in a turn.
5. Specific Objectives	11.1.3.1 Trainee will be able to describe flight performance of aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Analysis of turn forces using IB diagrams; Safety: No physical flight; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary forces acting on an aircraft in straight and level flight?
(b) How does the lift vector change when an aircraft banks?
(c) Define a coordinated turn in terms of ball position and slip/skid.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Centrifugal force in a turn • Load factor definition and calculation • Relationship between bank angle and load factor • Effect of load factor on stall speed • G-force sensation in high bank angles	L & Ds	PPT, IB	15 min
(b)	• Rate of turn formula derivation • Standard rate turn definition (3 degrees per second) • Bank angle required for standard rate at various speeds • Influence of airspeed on turn radius • Turn coordination and adverse yaw correction	L & Ds	PPT, PDF	20 min
(c)	• Control inputs for initiating a turn • Roll control via ailerons and rudder coordination • Pitch control to maintain altitude in a turn • Trim adjustments during sustained turns • Recovery from steep and shallow turns	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how increasing bank angle affects the stall speed of the aircraft.
(b) Describe the control inputs required to maintain a steady level coordinated turn.
(c) Calculate the bank angle needed for a standard rate turn at 90 knots.

12. Home Assignment and Follow Up	Prepare a brief note comparing the flight performance characteristics of a steep turn versus a shallow turn, referencing load factor and stall speed implications for the next lesson on stability.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Stability and control: static, dynamic stability. definition, notation used Stability requirements, a/c controls and Manoeuvres,	Refer to the master syllabus reference list	LP 22

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor Name, Instructor, P No. 12345	Senior Instructor, Senior Instructor, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min